

PRACTICE MIDTERM 1

Answer Key

CS 410/510 — Databases | Fall 2026

Work the practice exam under exam conditions before reading this. Where a question has more than one right answer, that is said so explicitly — on the real exam those alternatives earn full credit too.

Part A (15pts): Parts of an Entity-Relationship Diagram

Entity types (3)

SPEAKER, SESSION, TRACK, ROOM, ATTENDEE

Relationship types (3)

Presents, Registers_for, Belongs_to, Held_in

Primary keys (4)

SPEAKER: **SpeakerID** SESSION: **SessionID** TRACK: **TrackID** ROOM: **RoomID** ATTENDEE: **AttendeeID**

Foreign keys (5)

The two 1:N relationships put a foreign key on the many side:

- SESSION gets **TrackID** → TRACK (SESSION is the N side of Belongs_to)
- SESSION gets **RoomID** → ROOM (N side of Held_in)

The two M:N relationships cannot be folded into either side. Each becomes its own relation:

- **PRESENTS(SpeakerID → SPEAKER, SessionID → SESSION)**
- **REGISTRATION(AttendeeID → ATTENDEE, SessionID → SESSION)**

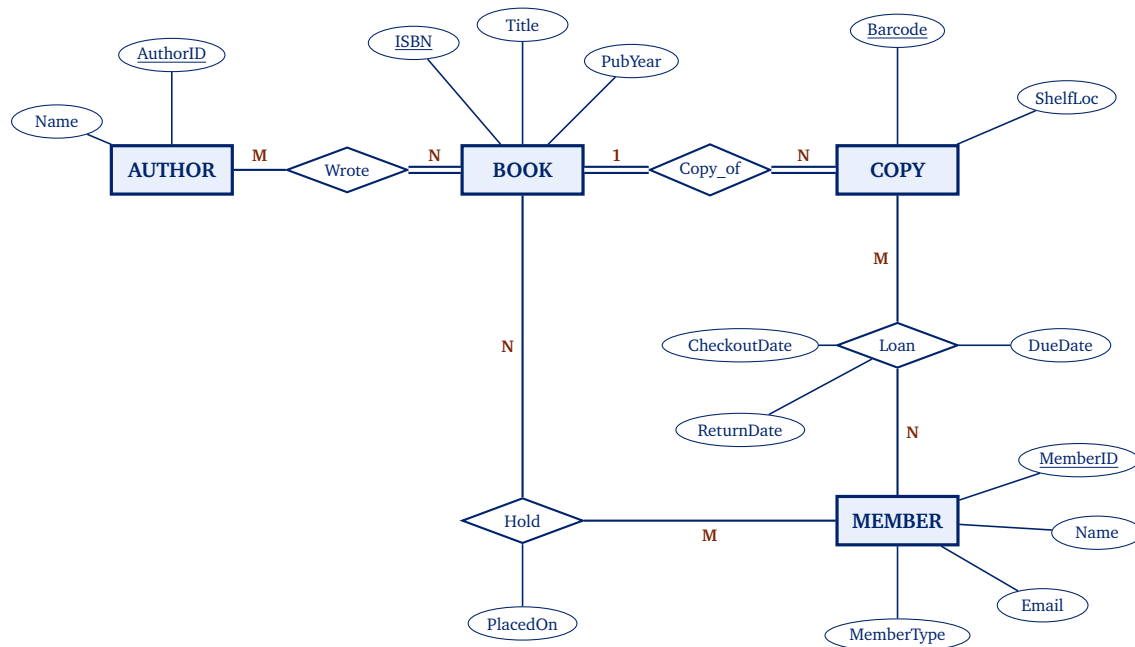
SPEAKER, TRACK, ROOM and ATTENDEE take **no foreign keys** of their own.

Why. This is the difference from the sample midterm, which had no M:N relationship. You cannot put **SpeakerID** on **SESSION** — a session has several speakers, and a column holds one value. The bridge relation is the only option, and in it *both* columns are foreign keys and *together* they are the primary key. That is the counterexample to “an attribute can be part of the primary key or a foreign key, but not both.”

Part B (40pts): Database Modeling — Public Library

ER Diagram (20)

One acceptable model:



Double line = total participation.

Why. The one thing this problem is really testing is that BOOK and COPY are different entities. BOOK is the *work* — it has the title, ISBN and authors. COPY is the *physical object* on the shelf — it has the barcode. A loan is of a COPY; a hold is on a BOOK, because you want whichever copy comes back first. Merge the two and you cannot represent “the library owns four copies,” and ISBN stops being a key.

Also graded: Wrote is M:N, since a book has one or more authors and an author writes many books; Loan carries attributes (the three dates) on the relationship; ReturnDate is nullable, which is how you tell an open loan from a closed one.

Equally correct: modeling Loan as its own entity rather than a relationship with attributes — it needs to be, if you want to keep a history of every past loan of the same copy by the same member. Also fine: COPY as a weak entity of BOOK with a copy number as partial key.

Relational Schema (20)

Underline = primary key. → = foreign key reference.

```

AUTHOR(AuthorID, Name)
BOOK(ISBN, Title, PubYear)
WROTE(AuthorID → AUTHOR, ISBN → BOOK)
COPY(Barcode, ISBN → BOOK, ShelfLoc)
MEMBER(MemberID, Name, Email, MemberType)
LOAN(LoanID, Barcode → COPY, MemberID → MEMBER,
      CheckoutDate, DueDate, ReturnDate)
HOLD(MemberID → MEMBER, ISBN → BOOK, PlacedOn)

```

Data types

Attribute	Type	Why
AuthorID, MemberID, LoanID	INTEGER	surrogate keys
PubYear	INTEGER	a year, not a date
ISBN, Barcode	VARCHAR	identifiers with leading zeros and dashes — not arithmetic
Title, Name, Email	VARCHAR	short bounded strings
ShelfLoc, MemberType	VARCHAR	or ENUM for MemberType
CheckoutDate, DueDate, ReturnDate, PlacedOn	DATE	a day is enough; use TIMESTAMP if the library tracks the hour

Why. ISBN as VARCHAR rather than a number is worth noticing: it can begin with a zero, can contain dashes, and ends in a check character that may be X. The test for a numeric type is whether you would ever do arithmetic on it. The same reasoning applies to a phone number or a ZIP code.

LOAN gets its own surrogate LoanID rather than (Barcode, MemberID) as a composite key — otherwise the same member could never borrow the same copy twice.

Part C (20pts): Writing Queries

1. Which artists formed in 1990 or later? (4)

```
SELECT * FROM Artist WHERE formed_year >= 1990;
```

RA: $\sigma_{\text{formed_year} \geq 1990}(\text{Artist})$

Returns: Nova Trio, Marisol Vega, Halvard

2. What are the titles of all jazz albums? (4)

```
SELECT title FROM Album WHERE genre = 'jazz';
```

RA: $\pi_{\text{title}}(\sigma_{\text{genre}='jazz'}(\text{Album}))$

Returns: Blue Arithmetic, Second Light

Why. `SELECT title, not SELECT *` — the question asks for titles. That is the projection.

3. What albums does The Quiet Hours have? (4)

```
SELECT b.* FROM Album b JOIN Artist a
  ON b.artist_id = a.artist_id WHERE a.name = 'The Quiet Hours';
```

RA: $\text{Album} \bowtie \sigma_{\text{name}='The Quiet Hours'}(\text{Artist})$

Returns: Longitude

4. Names of all artists who have a rock album? (4)

```
SELECT a.name FROM Artist a JOIN Album b
  ON a.artist_id = b.artist_id WHERE b.genre = 'rock';
```

RA: $\pi_{\text{Artist.name}}(\text{Artist} \bowtie \sigma_{\text{genre}='rock'}(\text{Album}))$

Returns: The Quiet Hours

Why. The join runs the other direction from question 3: filter on Album, project from Artist. Returning the album title here is the standard mistake.

5. How many albums does each artist have? (4)

```
SELECT a.name, COUNT(b.album_id) AS album_count
  FROM Artist a LEFT JOIN Album b ON a.artist_id = b.artist_id
 GROUP BY a.artist_id, a.name;
```

RA: $\text{artist_id} \mathcal{F}_{\text{COUNT}(\text{album_id})}(\text{Album})$

Returns: Nova Trio 2, The Quiet Hours 1, Marisol Vega 1, Halvard 0

Why. Halvard is the whole point of this question. Halvard has no albums, so an `INNER JOIN` drops the row entirely and you get three rows instead of four. “How many albums does *each* artist have” means every artist, including the ones with zero. `LEFT JOIN` keeps them, and `COUNT(b.album_id)` — counting the *album* column, not `COUNT(*)` — correctly yields 0 rather than 1, because `COUNT` of a column ignores `NULL`s.

Aggregation is outside basic relational algebra; the $\mathcal{F}$ notation above is the extended operator. Saying so is an acceptable answer.

Part D (15pts): Concepts — 1 point each

#	Ans	Why
1	F	Backwards. The schema is the structure and rarely changes; the instance is the data and changes constantly.
2	F	Reversed. Every candidate key is a superkey, but a superkey with a removable attribute — {StudentID, Name} — is not minimal, so it is not a candidate key.
3	T	Double oval = multivalued. Dashed oval = derived; double rectangle = weak entity.
4	T	Owner key + partial key gives the weak entity its full identity.
5	F	σ picks rows; π picks columns. If you missed this one, you likely also missed the mirrored item on the sample midterm.
6	F	Only M:N relationships always need their own relation. 1:1 and 1:N fold into a foreign key.
7	F	A derived attribute is computed on demand, not stored — that is the point of the dashed oval.
8	F	The foreign key goes on the <i>many</i> side. The one side cannot hold many references in a single column.
9	F	A primary key can never be NULL: entity integrity. A NULL cannot identify anything.
10	T	The natural join is a derived operator — product, then select on the matching columns, then project away the duplicate.
11	F	A column holds one value. A multivalued attribute becomes its own relation.
12	T	Double line = total participation; single line = partial.
13	T	Declarative, not procedural — you state the result you want and the DBMS plans the execution.
14	T	1:1, 1:N, N:1 and M:N can all map to the same three relations; only the key declarations differ. This is the Ch. 3 & 9 lecture.
15	F	A relation is a <i>set</i> of tuples — unordered. SQL tables do have an order, which is one of several ways SQL departs from the relational model.

Part E (10pts): Data Types — 1 point each

#	Item	Ans	Why
1	Seats in a lecture hall	A	INTEGER — a count.
2	A course catalog description	E	TEXT — a paragraph, not a label.
3	A student's photo ID image	F	BLOB — binary.
4	The exact price of a textbook	B	DECIMAL — money must be exact.
5	A building's street address	D	VARCHAR — short, variable-length.
6	The date a book is due	G	DATE — a day, no time.
7	When a payment was processed	I	TIMESTAMP — date and time together.
8	When a daily backup starts	H	TIME — a clock time, no date.
9	A measured humidity reading	C	REAL — a measurement; approximation is fine.
10	A department's abbreviation	D	VARCHAR — a short string.

Why. Choice D is correct twice. Do not assume each letter is used exactly once — that assumption forces a wrong answer somewhere else.

The pair to study is **4 vs. 9**. Both are “numbers with a decimal point,” but money needs DECIMAL because a rounding error of a hundredth of a cent compounds into real money, while a humidity sensor is only accurate to a fraction of a percent anyway, so REAL costs you nothing. Then **6 / 7 / 8** separate DATE, TIMESTAMP and TIME: ask whether you need the day, the moment, or the clock time.

If you want to check one thing before the exam, make it the mapping from a diagram to relations: which cardinality puts a foreign key where, and which one forces a new relation. Part A, Part B's schema, and Part D items 6, 8, 11 and 14 are all the same question asked five ways.